TASK 4

Final Data Analytics Project

SWYNEX Technologies Internship

1. Problem Statement

Startup funding data contains information about industries, funding stages, years, countries, employees and funding amounts. The objective of this final project is to organize and analyze the dataset to understand funding patterns and present the important findings through an interactive dashboard.

Goal: Clean the startup funding dataset, perform exploratory analysis, create meaningful visualizations, and summarize the key business insights in an easy-to-understand dashboard.

2. Dataset Information

Item	Details
Dataset	Startup Funding Dataset
Source	Public dataset
Records	Approximately 2,000 startup funding records
Main fields	Industry, Funding Stage, Funding Amount, Year, Country, Employees
Prepared data	Cleaned Excel Data Set from Task 1

3. Data Cleaning Process

- Used the cleaned Excel data set prepared in Task 1.
- Checked the dataset for duplicate records and removed duplicate entries.
- Checked for missing values and ensured the important fields were ready for analysis.
- Reviewed inconsistent or incorrect values and corrected them where required.
- Checked data types and formatting so the data could be used effectively for analysis and dashboard creation.

4. Data Analysis

The cleaned data was analyzed to identify patterns in startup funding across industries, funding stages, years, countries and employee counts.

Analysis	Purpose
Funding by Industry	Compare total funding across different industries.
Funding by Stage	Understand funding distribution across startup stages.
Funding by Year	Identify years with higher or lower total funding.

Average Funding by Country	Compare average startup funding across countries.
Funding vs Employees	Observe the relationship between funding and employee count.

5. Dashboard

- **KPI Cards:** Used to display important summary values such as total funding, average funding and record count.
- **Pie/Chart Visuals:** Used to show funding distribution by industry and funding stage.
- **Filters / Slicers:** Allow users to interact with the dashboard by selecting fields such as year, country, industry or funding stage.
- **Dashboard Layout:** Arranged KPI cards at the top, charts in the main analysis area and filters in an accessible section for easy interaction.
- **Interactive Analysis:** Selecting a filter or slicer updates the related dashboard visuals.

6. Key Findings

- AI was identified as the industry with the highest total funding in the analysis.
- Education had the highest number of startup records in the dataset.
- IPO was the funding stage with the highest total funding among the stages analyzed.
- Germany showed the highest average funding per startup among the countries included in the analysis.
- 2019 recorded the highest annual total funding in the analyzed dataset.
- The funding-versus-employees analysis showed a correlation of approximately -0.008, indicating almost no linear relationship in this dataset.

7. Business Insights

The analysis shows that startup funding varies considerably by industry, funding stage, year and country. The dashboard helps users quickly identify areas with higher funding and compare different categories using interactive filters. The employee comparison also suggests that employee count alone does not explain funding levels in this dataset.

8. Tools Used

- Excel – Data cleaning and preparation
- Power BI – Dashboard creation and interactive visual analysis

9. Project Outcome

The final project combines the work completed during the internship into a complete data analytics case study. The cleaned data was analyzed and converted into an interactive dashboard with KPIs, charts and slicers. The project provides a clear view of startup funding patterns and summarizes the main findings for decision-making.

10. Conclusion

This final project demonstrates the complete analytics workflow: understanding the problem, preparing the data, performing analysis, creating visualizations and communicating business insights. The interactive dashboard makes the results easier to explore and understand.